# Two Geometric Facts of the Sky, and the Observations Behind Them

Appendix — annotated literature list. (1) A star's altitude rises ~1° for every ~111 km travelled toward its geographical position; (2) the dip of the sea horizon follows a square-root law in observer height at low elevations. Compiled 2026.

## 1. One degree of altitude per ~111 km toward the geographical position (GP)

For a star, the incoming rays are effectively parallel, so the altitude measured at the observer and the great-circle angle subtended at Earth's centre between the observer and the sub-stellar point (GP) are equal. The zenith distance z = 90° – altitude therefore equals the angular ground distance from the GP, and moving 1° of arc toward the GP raises the star by 1°. Turning that 1° of arc into a ground distance requires Earth's size: 1° of great circle ~ 60 nautical miles ~ 111 km. The empirical support falls in four strands: the classical altitude-versus-distance measurement, the 17th–18th c. geodetic arc surveys that measured kilometres per degree of latitude directly from star altitudes, everyday celestial navigation, and the metrological definitions that fix the round number.

| Source | A posteriori evidence / relevance | Link |
|---|---|---|
| **Eratosthenes** (c. 240 BCE), transmitted by Cleomedes, *On the Heavens*. | First determination of Earth's size from the difference in the Sun's noon altitude between Syene and Alexandria and their ground separation — within ~2% of the modern value. The archetypal altitude/distance measurement. | NOAA |
| **Modern "Eratosthenes Experiment"** (schools worldwide, annual, equinox). | Repeatable classroom version: shadow angle vs. north–south distance recovers the km-per-degree ratio each year. Live demonstration of exactly this relation. | UT Austin |
| **J. Picard**, *Mesure de la Terre* (1671). Paris–Amiens meridian arc. | Early precise arc measurement: 1° of latitude = 57,060 toise, with latitude fixed by astronomical (zenith-sector) star observations. Foundation for the figure-of-the-Earth debate. | — |
| **French Geodesic Mission to Lapland** (Maupertuis, Clairaut, Camus, Outhier, 1736–37); Maupertuis, *La figure de la terre* (1738). | Measured a ~1° meridian arc (~111 km) in the Torne Valley (then Sweden), latitude from circumpolar stars via a zenith sector; proved the Earth is oblate. Direct km-per-degree measurement from star altitudes, at high latitude. | Wikipedia |
| **French Geodesic Mission to the Equator / Peru** (La Condamine, Bouguer, Godin, 1735–43); L. Ferreiro, *Measure of the Earth* (2011). | Companion equatorial arc: 1° of latitude = 110.61 km near the Equator. Comparison with the Lapland/Paris arcs quantified Earth's oblateness — i.e. showed the km-per-degree value is not perfectly constant. | Wikipedia |
| **Delambre & Méchain**, *Base du système métrique décimal* (1806–10); K. Alder, *The Measure of All Things* (2002). | Dunkirk–Barcelona meridian arc (1792–99) tied to astronomically-determined latitudes; used to define the metre as 1/10,000,000 of the pole-to-equator quadrant. The measurement that the "111 km" figure is ultimately calibrated to. | Wikipedia |
| **Bowditch**, *The American Practical Navigator* (NGA Pub. No. 9). Freely published by NGA. | Standard operational reference for the intercept (sight-reduction) method, where zenith distance is used as a ground distance from the GP at 60 nm per degree — confirmed by every sextant fix. | — |
| **Intercept method** (Marcq St. Hilaire, 1875), "Calcul du point observé." | The working principle that the zenith distance equals the GP's angular distance from the observer, with 1′ of altitude plotted as 1 nautical mile. Daily practical confirmation. | Wikipedia |
| **Nautical mile & metre definitions** (International Hydrographic Conf. 1929; 1791 metre). | Pins the exact number: 1 nautical mile = 1′ of latitude = 1,852 m, so 1° = 60 nm = 111.12 km. The round figure is downstream of the measured meridian arc above. | Wikipedia |

**Caveat (itself an a posteriori result).** Because Earth is an oblate spheroid, a degree of latitude is not constant: it runs from about 110.57 km near the Equator to about 111.69 km near the poles, so ~111.1–111.2 km is a mid-latitude / mean value. The geodetic arc surveys above are precisely the observations that revealed this variation.

**Scope.** The exactness for a *star* relies on parallel rays. For the Moon a parallax term is required, and near the horizon refraction shifts observed altitude — hence the second finding.

## 2. Dip of the sea horizon: a square-root law in eye height (low-elevation regime)

The apparent sea horizon sits below the astronomical horizon by an angle (the *dip*) that grows with the observer's eye height h. Geometry gives dip ~ sqrt(2h/R); in the low-elevation regime (h much smaller than Earth's radius R) this reduces to a square-root law. Two coefficients are in common use: the purely geometric value, dip[arcmin] ~ 1.93 × sqrt(h[m]) (equivalently 1.06 × sqrt(H[ft])); and the standard-refraction value adopted in the almanacs, dip[arcmin] ~ 1.76 × sqrt(h[m]) (equivalently 0.97 × sqrt(H[ft])). The corresponding distance to the horizon is ~3.57 × sqrt(h[m]) km geometrically, ~3.86 × sqrt(h[m]) km with standard refraction. The square-root form is a low-height approximation: at aircraft or mountain elevations the exact dip = arccos(R/(R+h)) departs from pure sqrt(h) and refraction variability grows. The a posteriori support is more than a century of direct shipboard dip measurements and their modern reanalysis.

| Source | A posteriori evidence / relevance | Link |
|---|---|---|
| **A. T. Young**, "Dip of the Horizon" and "Distance to the Horizon" (San Diego State University). | Authoritative modern treatment of the geometry plus terrestrial refraction; derives the sqrt(h) form, the standard 1.75–1.76′·sqrt(h[m]) coefficient, and the refraction factor k and its real-world variability. | SDSU |
| **H. C. Freiesleben** (1950), "Investigations into the Dip of the Horizon," *J. Navigation* 3(3), 270–279. DOI:10.1017/S037346330003486X. | States and analyses dip = 1.06·sqrt(H[ft]) = 1.93·sqrt(h[m]); reviews the historical shipboard measurement campaigns and the temperature-dependent anomalies around the standard law. | NavList PDF |
| **M. E. Tschudin** (2016), "Refraction near the horizon — an empirical approach. Part 1," *Applied Optics* 55(12), 3104–3115. DOI:10.1364/AO.55.003104 (erratum: *Appl. Opt.* 57, 3479, 2018). | Reanalyses over 1,800 previously published *direct measurements* of the sea horizon's depression; the derived one-parameter function agrees with the modern almanac and land-surveying sqrt(h) forms. The strongest single empirical validation. | Optica |
| **B. Chr. Peterson** (1952), "Some Observations of Refraction at Low Altitudes and of Astronomical Position-line Accuracy," *J. Navigation* 5, 31. | 1,908 sextant observations (made 1934) prompted by discussions with P. Collinder of the Swedish Hydrographic Service and K. Lundmark (Lund) — low-altitude refraction and horizon-elevation data. A Swedish contribution to the empirical record. | Cambridge |
| **Hessen, Huber & Thorade** (1930), *Kimmtiefenmessungen an Bord von Schiffen der Reichsmarine*, Veröff. Marine-Obs. Wilhelmshaven Nr. 15. | Systematic shipboard dip ("Kimmtiefe") measurements — one of the large historical datasets on which later analyses (Freiesleben; Tschudin) are built. | — |
| **E. Perrin** (1886), "Sur les dépressions de l'horizon de la mer," *C. R. Acad. Sci.* | Early quantitative measurements of sea-horizon depression, part of the observational chain feeding the standard dip formulae. | — |
| **D. H. Sadler & W. A. Scott**, "Refraction at Low Altitudes," *J. Navigation* (H.M. Nautical Almanac Office). | Analysis of many thousands of low-altitude observations concluding that the published almanac corrections (including the standard dip) need no significant amendment — operational validation of the sqrt(h) dip in use. | Cambridge |
| **Bowditch**, *The American Practical Navigator* (NGA Pub. No. 9); *Explanatory Supplement to the Astronomical Almanac*. | Canonical dip table and formula (0.97·sqrt(H[ft]) ~ 1.76·sqrt(h[m])), tabulated for practical eye heights and validated by generations of navigators. | — |
| **M. Hines**, "Correcting Sextant Measurements for Dip," Math Encounters (2015); cf. Starpath, *Distance to the Horizon*. | Worked derivations of the geometric and refraction-modified sqrt(h) dip formulae, cross-checked against Bowditch/Norie tables (agreement within a few percent, i.e. within atmospheric variation). | MathSciNotes |

**Why ‘low elevations’ matters.** The pure sqrt(h) law is the leading-order expansion of dip = arccos(R/(R+h)); it is accurate to well under a percent for eye heights up to tens of metres, but the exact form and refraction spread must be used for high observers. The shipboard datasets above sample precisely the low-height regime where the square-root law is expected to hold, which is why they test it cleanly.

**Refraction.** Terrestrial refraction reduces the geometric dip by roughly the factor (1 - k), with k ~ 1/7 under standard conditions but ranging from small negative values to greater than 1 over cold water — the dominant source of scatter in the measured dips.

Notes: dashes in the Link column mark books, primary sources, or historical papers without a stable open URL; cite by the bibliographic details given. Coefficient values are quoted to the precision commonly tabulated; small differences between sources reflect the choice of Earth radius and refraction constant.